



COMP30026

Models of Computation

Cezary Kaliszyk and William Umboh

Lecture 4

Resolution



Propositional Logic is Decidable

Valid/contingent/unsatisfiable? Decide with truth table.



Propositional Logic is Decidable

Valid/contingent/unsatisfiable? Decide with truth table.

Finite, but **huge!**

2^n rows where n is number of variables.



Either-Or Reasoning

If the password is correct, the user is logged in.
If the password is not correct, an error message is shown.

∴

Either-Or Reasoning

If the password is correct, the user is logged in.

If the password is not correct, an error message is shown.

$\therefore$ The user is either logged in or shown an error message.



Either-Or Reasoning, Symbolically

$$\frac{P \rightarrow F \quad \neg P \rightarrow G}{F \vee G}$$

$$\frac{P \rightarrow \perp \quad \neg P \rightarrow \perp}{\perp}$$

Rewrite “ $\rightarrow$ ” in terms of “ $\neg$ ” and “ $\vee$ ”:

$$\frac{\neg P \vee F \quad P \vee G}{F \vee G}$$

$$\frac{\neg P \quad P}{\perp}$$

This is **resolution**.

Resolution

Rewrite “ $\rightarrow$ ” in terms of “ $\neg$ ” and “ $\vee$ ”:

$$\frac{\neg P \vee F \quad P \vee G}{F \vee G}$$

$$\frac{\neg P \quad P}{\perp}$$

This is **resolution**.

Exercise: check this is sound!



Resolution Graph Example

Graphical representation of proof:

$$B \vee \neg A \quad B \vee A \quad \neg B$$

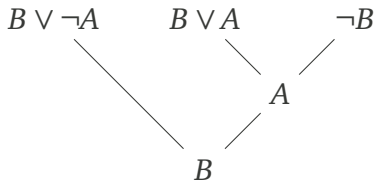
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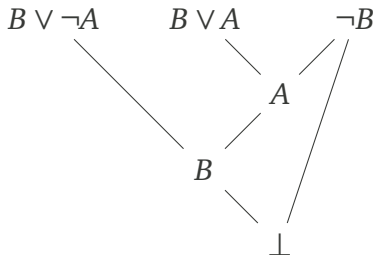
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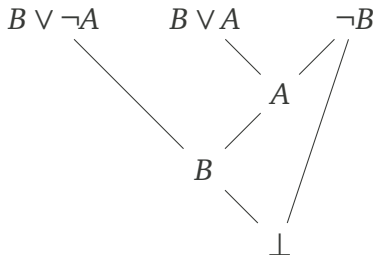
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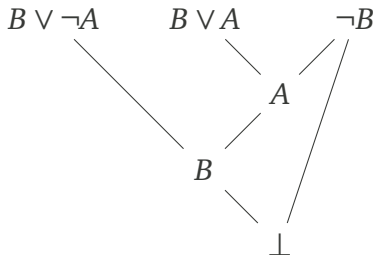
Graphical representation of proof:



A **refutation** of $(B \vee \neg A) \wedge (B \vee A) \wedge \neg B$

Resolution Graph Example

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Showing that it is not true; deriving contradiction



How to Use Refutations

To show that F is **valid**, refute $\neg F$.

Theorem

$\models F$ iff $\neg F \models \perp$.

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To prove $F \models G$, refute $F \wedge \neg G$.

Theorem

$F \models G$ iff $F \wedge \neg G \models \perp$.



Resolution Proof Exercise

Refute:

$$P \vee \neg Q \quad \neg P \quad Q \vee \neg R \quad Q \vee R \vee S \quad \neg S$$

Resolution Proof Exercise

Refute:

$$\begin{array}{ccc} P \vee \neg Q & & \neg P \\ & \diagdown \quad \diagup & \\ & \neg Q & \end{array}$$

$$\begin{array}{ccccc} Q \vee \neg R & & Q \vee R \vee S & & \neg S \\ & & \diagdown \quad \diagup & & \\ & & Q \vee R & & \end{array}$$

Resolution Proof Exercise

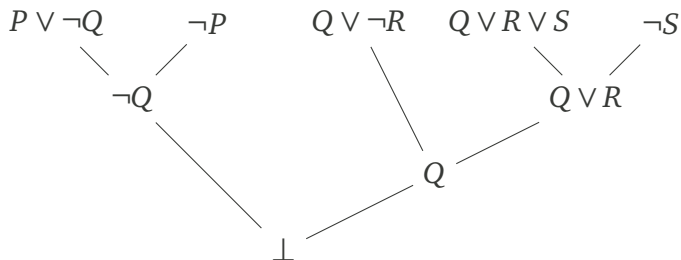
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Resolution Proof Exercise

Refute:





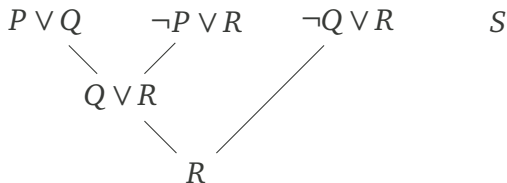
Resolution Proof Exercise

Derive R from the premises:

$$P \vee Q \quad \neg P \vee R \quad \neg Q \vee R \quad S$$

Resolution Proof Exercise

Derive R from the premises:





Do Not Do This, Please

Do not cancel *multiple* variables at once!



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$(P \vee Q) \wedge (\neg P \vee \neg Q)$ is **satisfiable!!!**

If you could cancel both, you would get $\perp$!!!



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$(P \vee Q) \wedge (\neg P \vee \neg Q)$ is **satisfiable!!!**

If you could cancel both, you would get $\perp$!!!

This would not be sound!!!

Propositional Resolution Formally

Definition

A resolution proof of C_m from wffs $P_1, \dots, P_n$ is a **string** of the form

$$P_1, \dots, P_n \vdash C_1, \dots, C_m$$

where each C_i is either a **copy** of some P_j , or otherwise follows by **resolution** from any two wffs **earlier in the string**.

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where each C_i is either a **copy** of some P_j , or otherwise follows by **resolution** from any two wffs **earlier in the string**.

Examples:

- “ $P \vdash P$ ”
- “ $P, \neg P \vdash \perp$ ”
- “ $(P \vee Q), \neg P \vdash Q$ ”



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We write “ $\Sigma \vdash_R F$ ” to mean “there is a resolution proof of F from the set of premises Σ ”.



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Proof (sketch).

Let x be a proof of F from Σ .

Let v be a model of Σ .

Let $C_1, \dots, C_n$ be the wffs after the $\vdash$ in x .

Prove by induction that v satisfies each C_i .





Conjunctive Normal Form (CNF)

Literal: a propositional atom or its negation.

(Disjunctive) clause: disjunction ($\vee$) of literals.

CNF: conjunction ($\wedge$) of disjunctive clauses.



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$$(A \vee \neg B) \wedge (B \vee C \vee D) \wedge A$$



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Theorem

Every propositional formula has an equivalent CNF.



Negation Normal Form (NNF) to CNF

NNF: Only connectives are $\neg$, $\wedge$ and $\vee$.
 $\neg$ only in front of variables.

Example

$$(\neg A \vee (B \wedge \neg C)) \vee (C \wedge (B \vee D))$$



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To get NNF:

1. Eliminate $\leftrightarrow$ (rewrite using $\rightarrow$ and $\wedge$).
2. Eliminate $\rightarrow$ (rewrite using $\vee$ and $\neg$).
3. Push $\neg$ inward (use de Morgan's laws).
4. Eliminate $\neg\neg$.

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To get CNF from NNF, distribute $\vee$ over $\wedge$.



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$$(\neg P \wedge (\neg Q \rightarrow R)) \leftrightarrow S$$



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$$\begin{aligned} & (\neg P \wedge (\neg Q \rightarrow R)) \leftrightarrow S \\ \equiv & ((\neg P \wedge (\neg Q \rightarrow R)) \rightarrow S) \wedge (S \rightarrow (\neg P \wedge (\neg Q \rightarrow R))) \end{aligned} \quad (1)$$

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$$\equiv ((P \vee (\neg Q \wedge \neg R)) \vee S) \wedge (\neg S \vee (\neg P \wedge (Q \vee R))) \quad (4)$$

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Example Conversion to CNF

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 \end{aligned}$$

The result is in conjunctive normal form.

Resolution is Refutation-Complete

Theorem

Every unsatisfiable set of clauses has a resolution refutation.

In other words:

Theorem

Let Σ be a set of clauses.

If $\Sigma \models \perp$, then $\Sigma \vdash_R \perp$.



Satisfiability Algorithm

This gives us an algorithm:

1. Convert formula into suitable form.
2. Repeatedly apply resolution.
 - Derive $\perp$? Report **unsatisfiable**.
 - Cannot derive anything new? Report **satisfiable**.



Simplifying CNF

Common redundancies:

- Duplicate variables (e.g. $P \vee P$)
- Tautologies (e.g. $P \vee \neg P \vee Q$)
- Subsumptions (e.g. $(P \vee \neg Q \vee R) \wedge (P \vee R)$)

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Exercise: simplify this formula:

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Note that CNF is **not** unique!

Clausal Form

Represent CNF as set of sets of literals.

Example

CNF:

$$(P \vee \neg Q \vee S) \wedge (P \vee \neg R \vee S) \wedge (\neg S \vee \neg P) \wedge (\neg S \vee Q \vee R)$$

Clausal form:

$$\{\{P, S, \neg Q\}, \{P, S, \neg R\}, \{\neg P, \neg S\}, \{Q, R, \neg S\}\}$$

We shall often treat these interchangeably.

Why? Simplifies reasoning.



Empty Disjunction

Let A and B be propositional variables.

- $\{A, B\}$ represents the clause $A \vee B$.
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Natural choice: $\perp$.



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Natural choice: $\perp$.

Disjunction is true iff **at least one** disjunct is true.



Empty Conjunction

Let C_1 and C_2 be clauses.

- $\{C_1, C_2\}$ represents the CNF formula $C_1 \wedge C_2$.
- $\{C_1\}$ represents the CNF C_1 .

What **CNF** does $\{\}$ represent?



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What **CNF** does $\{\}$ represent?

Natural choice: $\top$.

Conjunction is true iff **every** conjunct is true.



Empty Clauses and Formulas

- Empty conjunction ($\wedge$) is **valid**.
- Empty disjunction ($\vee$) is **unsatisfiable**.



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Thus:

- The empty set of clauses is **valid**.
- A set of clauses containing the empty clause is **unsatisfiable**.



Propositional Resolution for Clausal Form

Let P be a propositional variable.

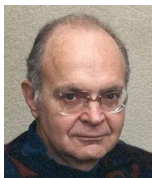
Let C_1, C_2 be clauses without P or $\neg P$.

$$\frac{C_1 \cup \{P\} \quad C_2 \cup \{\neg P\}}{C_1 \cup C_2}$$



“SAT is a key technology of the 21st century.”

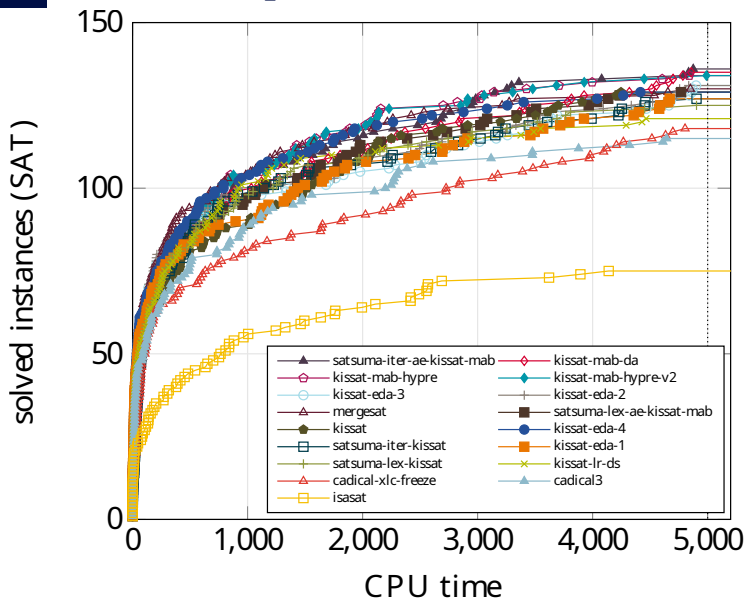
-E. Clarke, Handbook of Satisfiability



“The SAT problem is evidently a ‘killer app’ because it is key to the solution of so many other problems.”

-D. Knuth, Art of Computer Programming

SAT Competition 2026





Summary

So far

- Encoding “hard” problems as propositional logic
- If possible in CNF
- Resolution (or modern SAT solvers) can decide them

Next

- Even harder problems
- Predicate logic